

Nonlinear Optimization: Homework #4

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1 Duality

1st Problem

70. Binary classifier. Consider a finite set $\mathcal{Y} = \{y_1, \dots, y_K\}$. A probability mass function (pmf) on $\mathcal{Y}$ is a function $\mathbf{P}: \mathcal{Y} \rightarrow \mathbf{R}$ that assigns a probability to each point in $\mathcal{Y}$. Thus, $\mathbf{P}$ is nonnegative (which means $\mathbf{P}(y_k) \geq 0$ for each $1 \leq k \leq K$) and sums to one (which means $\mathbf{P}(y_1) + \dots + \mathbf{P}(y_K) = 1$).

Let $\mathbf{P}_0$ and $\mathbf{P}_1$ be two given pmfs on $\mathcal{Y}$ that are hypothesized for a random variable Y that takes values in $\mathcal{Y}$. In this problem, we address the following question: we observe a realization of the random variable Y and, on the basis of this observation, we want to decide if Y is following the pmf $\mathbf{P}_0$ or the pmf $\mathbf{P}_1$.

This question—known as classification in machine learning, or detection in signal processing, or hypothesis testing in statistics—is often denoted as

$$\begin{aligned} H_0 : Y &\sim \mathbf{P}_0, \\ H_1 : Y &\sim \mathbf{P}_1, \end{aligned}$$

with H_0 and H_1 representing the two hypotheses (or ground truths) under consideration.

Answering this question requires us to assign either the answer 0 or the answer 1 to each point in $\mathcal{Y}$: we assign the answer 0 to y_k if we believe y_k was drawn from $\mathbf{P}_0$, and we assign the answer 1 otherwise. In sum, answering the question requires us to design a map $\delta: \mathcal{Y} \rightarrow \{0, 1\}$, commonly called a classifier.

In our setting, a classifier δ can be represented by a K -dimensional binary vector

$$\delta = \begin{bmatrix} \delta_1 \\ \vdots \\ \delta_K \end{bmatrix},$$

where $\delta_k \in \{0, 1\}$ is the output of the classifier when the random variable Y is observed to be $y_k \in \mathcal{Y}$.

Thus, there are many classifiers as binary vectors of size K , that is, 2^K . Which one is the best? Recall that the role of a classifier is to guess if the underlying ground truth is H_0 or H_1 , on the basis of the random variable Y . Thus, a perfect classifier δ has two properties: $\delta(Y) = 0$ whenever Y is drawn from $\mathbf{P}_0$; and $\delta(Y) = 1$ whenever Y is drawn from $\mathbf{P}_1$. In vector notation, these two properties correspond to

$$\mathbf{P}_0 \delta = 0, \quad \mathbf{P}_1 \delta = 1,$$

where $\mathbf{P}_0$ and $\mathbf{P}_1$ are row vectors that represent the pmfs $\mathbf{P}_0$ and $\mathbf{P}_1$:

$$\mathbf{P}_0 = [\mathbf{P}_0(y_1) \quad \dots \quad \mathbf{P}_0(y_K)], \quad \mathbf{P}_1 = [\mathbf{P}_1(y_1) \quad \dots \quad \mathbf{P}_1(y_K)].$$

Indeed, note that for any classifier δ and any $h \in \{0, 1\}$, the quantity

$$\mathbf{P}_h \delta = [\mathbf{P}_h(y_1) \quad \cdots \quad \mathbf{P}_h(y_K)] \begin{bmatrix} \delta_1 \\ \vdots \\ \delta_K \end{bmatrix} = \sum_{k=1}^K \mathbf{P}_h(y_k) \delta_k$$

gives the probability of $\delta(Y) = 1$ when Y follows the distribution $\mathbf{P}_h$. The quantity $\mathbf{P}_1 \delta$ is commonly called the probability of detection, and the quantity $\mathbf{P}_0 \delta$ is commonly called the probability of false alarm (this is the probability of $\delta(Y) = 1$ when Y follows the distribution $\mathbf{P}_0$). Thus, a perfect classifier has probability of detection 1 and probability of false alarm 0, which is another way to interpret the properties above.

Unfortunately, a perfect classifier seldom exists. It exists only when the pmfs $\mathbf{P}_0$ and $\mathbf{P}_1$ have disjoint support. For example, if

$$\mathbf{P}_0 = [0.1 \quad 0.8 \quad 0.1 \quad 0 \quad 0] \quad \mathbf{P}_1 = [0 \quad 0 \quad 0 \quad 0.7 \quad 0.3],$$

a perfect classifier exists:

$$\delta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

As soon as the pmfs $\mathbf{P}_0$ and $\mathbf{P}_1$ overlap, a perfect classifier cannot exist. This is the case in most applications. In those cases, it is common to look for a classifier that maximizes the probability of detection, while keeping the probability of false alarm below a given tolerance level α (chosen in the open interval $]0, 1[$):

$$\begin{aligned} & \underset{\delta}{\text{maximize}} && \mathbf{P}_1 \delta \\ & \text{subject to} && \mathbf{P}_0 \delta \leq \alpha, \\ & && \delta_k \in \{0, 1\}, \quad k = 1, \dots, K. \end{aligned}$$

However, this problem is hard to solve due to the integer constraints $\delta_k \in \{0, 1\}$ imposed on the components of the vector δ . A natural approximation is to relax these integer constraints and allow δ_k to take values in the interval $[0, 1]$, which yields the following linear program:

$$\begin{aligned} & \underset{\delta}{\text{maximize}} && \mathbf{P}_1 \delta \\ & \text{subject to} && \mathbf{P}_0 \delta \leq \alpha, \\ & && 0 \leq \delta, \\ & && \delta \leq \mathbf{1}. \end{aligned}$$

Show that the dual problem of the relaxed problem can be written as

$$\begin{aligned} & \underset{\lambda}{\text{minimize}} && (\mathbf{P}_1 - \lambda \mathbf{P}_0)_+ \mathbf{1} + \lambda \alpha \\ & \text{subject to} && \lambda \geq 0. \end{aligned}$$

Solution to the 1st Problem. To simplify the solution, we start by folding the $2K$ linear constraints $0 \leq \delta \leq \mathbf{1}_K \iff \delta \in [0, 1]^K$ into the domain (which is equivalent to subtracting an indicator function over that set from the objective). The equivalent primal then reads

$$\begin{aligned} \text{(P)} \quad & \underset{\delta \in [0, 1]^K}{\text{maximize}} && \mathbf{P}_1 \delta \\ & \text{subject to} && \mathbf{P}_0 \delta - \alpha \leq 0, \end{aligned} \tag{1}$$

a linear program carrying a single explicit (affine) inequality constraint. We derive the dual in three steps, associating a scalar multiplier $\lambda \in \mathbf{R}_+$ to the constraint $\mathbf{P}_0 \delta - \alpha \leq 0$ as follows.

1) **Lagrangian.** Working directly with the maximization, we subtract the constraint term (keeping the Lagrangian above the objective at every feasible point). This is,

$$L : [0, 1]^K \times \mathbf{R}_+ \rightarrow \mathbf{R}, \quad L(\delta; \lambda) = \mathbf{P}_1 \delta - \lambda(\mathbf{P}_0 \delta - \alpha) = \lambda \alpha + (\mathbf{P}_1 - \lambda \mathbf{P}_0) \delta. \quad (2)$$

2) **Dual function.** For a maximization, the dual function is the supremum of L over the domain $\delta \in [0, 1]^K$. Since L is affine in δ and the domain decouples across coordinates, the supremum separates component-wise, this is,

$$g : \mathbf{R}_+ \rightarrow \mathbf{R}, \quad g(\lambda) = \sup \{L(\delta; \lambda) : \delta \in [0, 1]^K\} = \lambda \alpha + \sum_{k=1}^K \sup \{c_k \delta_k : 0 \leq \delta_k \leq 1\}. \quad (3)$$

where we defined $c_k := \mathbf{P}_1(y_k) - \lambda \mathbf{P}_0(y_k)$, for $k = 1, \dots, K$.

To evaluate each scalar supremum, we use the fact that, for $c \in \mathbf{R}$, maximizing a linear function over the unit interval admits the closed form $\sup\{c t : 0 \leq t \leq 1\} = \max(0, c)$. To see this, notice the map $t \mapsto c t$ is monotone on $[0, 1]$, so its maximum is always achieved at one of the endpoints:

- If $c \geq 0$, the map is nondecreasing, so the maximum is achieved at $t = 1$, with value c ;
- If $c < 0$, the map is decreasing, so the maximum is achieved at $t = 0$, with value 0.

In either case, $\sup\{c t : 0 \leq t \leq 1\} = \max(0, c) = (c)_+$. Applying this identity to each term in (3) gives

$$\sup\{c_k \delta_k : 0 \leq \delta_k \leq 1\} = (\mathbf{P}_1(y_k) - \lambda \mathbf{P}_0(y_k))_+,$$

and therefore, summing over k ,

$$g(\lambda) = \lambda \alpha + \sum_{k=1}^K (\mathbf{P}_1(y_k) - \lambda \mathbf{P}_0(y_k))_+ = \lambda \alpha + (\mathbf{P}_1 - \lambda \mathbf{P}_0)_+ \mathbf{1}_K, \quad (4)$$

where $(\cdot)_+$ denotes the entry-wise positive part, acting on the row vector $\mathbf{P}_1 - \lambda \mathbf{P}_0$.

3) **Dual problem.** As g upper-bounds the primal optimum, the dual problem minimizes this bound over $\lambda \in \mathbf{R}_+$,

$$\begin{aligned} \text{(D)} \quad & \underset{\lambda}{\text{minimize}} \quad (\mathbf{P}_1 - \lambda \mathbf{P}_0)_+ \mathbf{1}_K + \lambda \alpha \\ & \text{subject to} \quad \lambda \geq 0. \end{aligned} \quad (5)$$

The problem in (5) is therefore the same dual of the relaxed linear program as the one displayed in the problem statement. ☺

On the Strong Duality of the 1st Problem. We now discuss strong duality. Let p^* and d^* denote the optimal values of the relaxed primal (1) and its dual (5), respectively. The relaxed primal is an LP as its objective $\mathbf{P}_1 \delta$ is linear (hence convex) and its constraints $\mathbf{P}_0 \delta \leq \alpha$ and $0 \leq \delta \leq \mathbf{1}_K$ are affine (hence convex). Since the constraints are affine, a Slater point requires no strict inequality and feasibility suffices. Such a point exists (e.g. $\bar{\delta} = 0$), so

$$\exists \bar{\delta} : \mathbf{P}_0 \bar{\delta} \leq \alpha, \quad 0 \leq \bar{\delta} \leq \mathbf{1}_K \quad (\text{Slater point}).$$

The objective is also bounded above on the feasible set. Since $\mathbf{P}_1 \geq 0$ and $\delta \leq \mathbf{1}_K$,

$$\mathbf{P}_1 \delta \leq \mathbf{P}_1 \mathbf{1}_K = 1,$$

the last equality because $\mathbf{P}_1$ is a pmf. Hence $p^* \leq 1$. Since the primal is feasible, $p^* \in \mathbf{R}$. Then, Theorem (Strong Duality) implies

$$p^* = d^*,$$

and the dual is also solvable since $p^* \in \mathbf{R}$. ☺

2nd Problem

75. Dualizing a problem without constraints. Consider the optimization problem

$$\underset{x}{\text{minimize}} \quad \frac{1}{2}\|y - Ax\|^2 + \sum_{k=1}^K \rho_k \|G_k x\|$$

where y, A, G_k and $\rho_k > 0$ are given. (This problem structure arises in group sparse regularization applications.)

The goal of this problem is to dualize this problem. There are no constraints in the above to dualize, but we can consider the reformulation

$$\begin{aligned} \underset{x, z, w_1, \dots, w_K}{\text{minimize}} \quad & \frac{1}{2}\|y - z\|^2 + \sum_{k=1}^K \rho_k \|w_k\| \\ \text{subject to} \quad & z = Ax, \\ & w_k = G_k x, \quad k = 1, \dots, K. \end{aligned}$$

Show that the dual of the problem above can be written as

$$\begin{aligned} \underset{\lambda, \eta_1, \dots, \eta_K}{\text{maximize}} \quad & -\frac{1}{2}\|\lambda - y\|^2 + \frac{1}{2}\|y\|^2 \\ \text{subject to} \quad & A^T \lambda = \sum_{k=1}^K G_k^T \eta_k, \\ & \|\eta_k\| \leq \rho_k, \quad k = 1, \dots, K. \end{aligned}$$

Hint: the fact

$$\sup \{s^T x - \|x\| : x\} = \begin{cases} 0, & \text{if } \|s\| \leq 1, \\ +\infty, & \text{otherwise,} \end{cases}$$

may be helpful (if you use this fact, you must also prove it).

Solution to the 2nd Problem. We start from the equivalent constrained formulation

$$\begin{aligned} \text{(P)} \quad \underset{x, z, w}{\text{minimize}} \quad & \frac{1}{2}\|y - z\|^2 + \sum_{k=1}^K \rho_k \|w_k\| \\ \text{subject to} \quad & z - Ax = 0, \\ & G_k x - w_k = 0, \quad k = 1, \dots, K. \end{aligned} \tag{6}$$

where $w := (w_1, \dots, w_K)$ was used to aggregate the K variables.

Throughout, we assume $A \in \mathbf{R}^{m \times n}$ and $G_k \in \mathbf{R}^{m_k \times n}$, so $x \in \mathbf{R}^n$, $y, z \in \mathbf{R}^m$, and $w_k \in \mathbf{R}^{m_k}$. We derive the dual in the usual steps, associating the multiplier $\lambda \in \mathbf{R}^m$ to the constraint $z - Ax = 0$ and the multipliers $\eta_k \in \mathbf{R}^{m_k}$ to the constraints $G_k x - w_k = 0$, for $k = 1, \dots, K$. In this case, since all constraints are equalities, the multipliers are free. To simplify the notation, define

$$\mathcal{W} := \mathbf{R}^{m_1} \times \dots \times \mathbf{R}^{m_K}, \quad \eta := (\eta_1, \dots, \eta_K).$$

1) Lagrangian. Let $L : \mathbf{R}^n \times \mathbf{R}^m \times \mathcal{W} \times \mathbf{R}^m \times \mathcal{W} \rightarrow \mathbf{R}$,

$$L(x, z, w; \lambda, \eta) = \frac{1}{2}\|y - z\|^2 + \sum_{k=1}^K \rho_k \|w_k\| + \lambda^T (z - Ax) + \sum_{k=1}^K \eta_k^T (G_k x - w_k). \tag{7}$$

Rearranging some terms according to the primal variables gives

$$L(x, z, w; \lambda, \eta) = \frac{1}{2} \|y - z\|^2 + \lambda^T z + \sum_{k=1}^K (\rho_k \|w_k\| - \eta_k^T w_k) + \left(\sum_{k=1}^K G_k^T \eta_k - A^T \lambda \right)^T x. \quad (8)$$

2) Dual function. For a minimization, the dual function is the infimum of L over all the primal variables,

$$g : \mathbf{R}^m \times \mathcal{W} \rightarrow \mathbf{R} \cup \{-\infty\}, \quad g(\lambda, \eta) = \inf \{L(x, z, w; \lambda, \eta) : x \in \mathbf{R}^n, z \in \mathbf{R}^m, w \in \mathcal{W}\}. \quad (9)$$

From (8), the infimum separates into three parts.

- First, for the term depending on z , completing the square gives

$$\frac{1}{2} \|y - z\|^2 + \lambda^T z = \frac{1}{2} \|z - (y - \lambda)\|^2 - \frac{1}{2} \|\lambda - y\|^2 + \frac{1}{2} \|y\|^2.$$

Therefore,

$$\inf \left\{ \frac{1}{2} \|y - z\|^2 + \lambda^T z : z \in \mathbf{R}^m \right\} = -\frac{1}{2} \|\lambda - y\|^2 + \frac{1}{2} \|y\|^2. \quad (10)$$

- Second, we prove the fact stated in the hint. For any $s \in \mathbf{R}^q$,

$$\sup \{s^T u - \|u\| : u \in \mathbf{R}^q\} = \begin{cases} 0, & \text{if } \|s\| \leq 1, \\ +\infty, & \text{otherwise.} \end{cases}$$

This can be proven as follows.

- If $\|s\| \leq 1$, then by the Cauchy-Schwarz Inequality,

$$s^T u - \|u\| \leq \|s\| \|u\| - \|u\| = (\|s\| - 1) \|u\| \leq 0,$$

and equality is achieved for $u = 0$.

- If $\|s\| > 1$, choosing $u = t s / \|s\|$, with $t \geq 0$, gives

$$s^T u - \|u\| = t(\|s\| - 1) \rightarrow +\infty.$$

Applying this fact with $s = \eta_k / \rho_k$, and using $\rho_k > 0$, gives

$$\begin{aligned} \inf \{ \rho_k \|w_k\| - \eta_k^T w_k : w_k \in \mathbf{R}^{m_k} \} &= -\rho_k \sup \left\{ \left(\frac{\eta_k}{\rho_k} \right)^T u - \|u\| : u \in \mathbf{R}^{m_k} \right\} \\ &= \begin{cases} 0, & \text{if } \|\eta_k\| \leq \rho_k, \\ -\infty, & \text{otherwise.} \end{cases} \end{aligned} \quad (11)$$

- Finally, for the term depending on x , we have

$$\inf \left\{ \left(\sum_{k=1}^K G_k^T \eta_k - A^T \lambda \right)^T x : x \in \mathbf{R}^n \right\} = \begin{cases} 0, & \text{if } A^T \lambda = \sum_{k=1}^K G_k^T \eta_k, \\ -\infty, & \text{otherwise,} \end{cases} \quad (12)$$

because a nonzero linear function is unbounded below on $\mathbf{R}^n$.

Combining (10), (11) and (12), we obtain

$$g(\lambda, \eta) = \begin{cases} -\frac{1}{2} \|\lambda - y\|^2 + \frac{1}{2} \|y\|^2, & \text{if } A^T \lambda = \sum_{k=1}^K G_k^T \eta_k \wedge \|\eta_k\| \leq \rho_k, \quad k = 1, \dots, K, \\ -\infty, & \text{otherwise.} \end{cases} \quad (13)$$

3) **Dual problem.** As g lower-bounds the primal optimum, the dual problem maximizes this bound over $(\lambda, \eta) \in \mathbf{R}^m \times \mathcal{W}$. Therefore,

$$\begin{aligned}
 \text{(D)} \quad & \underset{\lambda, \eta}{\text{maximize}} \quad -\frac{1}{2}\|\lambda - y\|^2 + \frac{1}{2}\|y\|^2 \\
 & \text{subject to} \quad A^T \lambda = \sum_{k=1}^K G_k^T \eta_k, \\
 & \quad \|\eta_k\| \leq \rho_k, \quad k = 1, \dots, K.
 \end{aligned} \tag{14}$$

The problem in (14) is therefore the same dual problem as the one displayed in the statement. ☺

On the Strong Duality of the 2nd Problem. We now discuss the strong duality. Let p^* and d^* denote the optimal values of the primal (6) and its dual (14), respectively. The primal is convex as its objective $\frac{1}{2}\|y - z\|^2 + \sum_{k=1}^K \rho_k \|w_k\|$ is a squared norm plus a sum of norms (hence convex) and its constraints $z = Ax$ and $G_k x = w_k$, $k = 1, \dots, K$, are affine (hence convex) equalities. Since the constraints are affine equalities, a feasible point is a Slater point. Such a point exists (e.g. $(\bar{x}, \bar{z}, \bar{w}) = (0, 0, 0)$), so

$$\exists (\bar{x}, \bar{z}, \bar{w}) : \bar{z} = A\bar{x}, \bar{w}_k = G_k \bar{x}, \quad k = 1, \dots, K \quad (\text{Slater point}).$$

The objective is nonnegative and equals $\frac{1}{2}\|y\|^2$ at $(0, 0, 0)$, so $0 \leq p^* \leq \frac{1}{2}\|y\|^2$ and $p^* \in \mathbf{R}$. Theorem (Strong Duality) therefore implies that

$$p^* = d^*,$$

and the dual is also solvable since $p^* \in \mathbf{R}$. Since the reformulation (6) is equivalent to the original problem, the dual (14) dualizes the original problem with no duality gap. ☺

3rd Problem

77. Separating a point from a polyhedron. The distance from a point $p \in \mathbf{R}^n$ to a given, non-empty polyhedron $\mathcal{P} = \{x \in \mathbf{R}^n : Ax \leq b\}$ with $A \in \mathbf{R}^{m \times n}$ and $b \in \mathbf{R}^m$ is the optimal value of the optimization problem

$$\begin{aligned}
 & \underset{x \in \mathbf{R}^n}{\text{minimize}} \quad \|x - p\| \\
 & \text{subject to} \quad Ax \leq b.
 \end{aligned}$$

(a) Show that the dual problem is given by

$$\begin{aligned}
 & \underset{\mu \in \mathbf{R}^m}{\text{maximize}} \quad \mu^T (Ap - b) \\
 & \text{subject to} \quad \|A^T \mu\| \leq 1, \\
 & \quad \mu \geq 0.
 \end{aligned}$$

(b) Suppose the point p doesn't belong to the polyhedron $\mathcal{P}$, i.e., $p \notin \mathcal{P}$. Use the dual problem to show that there exists a hyperplane that separates p from $\mathcal{P}$. That is, show that there exists a nonzero $s \in \mathbf{R}^n$ and a $r \in \mathbf{R}$ such that $s^T p > r$ and $s^T x \leq r$ for all $x \in \mathcal{P}$.

Hint: if $p \notin \mathcal{P}$, the optimal value of the dual problem is positive.

Solution to the 3rd Problem. The solution is divided into two parts as follows.

(a) We start from

$$\begin{aligned}
 \text{(P)} \quad & \underset{x}{\text{minimize}} \quad \|x - p\| \\
 & \text{subject to} \quad Ax - b \leq 0.
 \end{aligned} \tag{15}$$

We derive the dual in three steps, associating a multiplier $\mu \in \mathbf{R}_+^m$ with the constraints $Ax - b \leq 0$.

1) **Lagrangian.** Since the primal is a minimization problem, we add the constraint term. This is,

$$L : \mathbf{R}^n \times \mathbf{R}_+^m \rightarrow \mathbf{R}, \quad L(x; \mu) = \|x - p\| + \mu^T (Ax - b). \quad (16)$$

2) **Dual function.** For a minimization, the dual function is the infimum of L over the primal variable:

$$g : \mathbf{R}_+^m \rightarrow \mathbf{R} \cup \{-\infty\}, \quad g(\mu) = \inf \{L(x; \mu) : x \in \mathbf{R}^n\}. \quad (17)$$

To compute this infimum, we define $u = x - p$, or equivalently $x = p + u$. Then

$$\begin{aligned} g(\mu) &= \inf \{\|u\| + \mu^T (A(p + u) - b) : u \in \mathbf{R}^n\} \\ &= \mu^T (Ap - b) + \inf \{\|u\| + (A^T \mu)^T u : u \in \mathbf{R}^n\}. \end{aligned} \quad (18)$$

Next, we use the fact that, for any $s \in \mathbf{R}^n$,

$$\inf \{\|u\| + s^T u : u \in \mathbf{R}^n\} = \begin{cases} 0, & \text{if } \|s\| \leq 1, \\ -\infty, & \text{otherwise.} \end{cases}$$

This can be proven as follows.

► If $\|s\| > 1$, taking $u = -t s / \|s\|$ gives

$$\|u\| + s^T u = t - t\|s\| = -t(\|s\| - 1) \rightarrow -\infty.$$

► If $\|s\| \leq 1$, then by the Cauchy-Schwarz Inequality,

$$\|u\| + s^T u \geq \|u\| - \|s\| \|u\| = (1 - \|s\|)\|u\| \geq 0,$$

with equality at $u = 0$.

Applying this fact with $s = A^T \mu$ gives

$$g(\mu) = \begin{cases} \mu^T (Ap - b), & \text{if } \|A^T \mu\| \leq 1, \\ -\infty, & \text{otherwise.} \end{cases} \quad (19)$$

3) **Dual problem.** As g lower-bounds the primal optimum, the dual problem maximizes this bound over $\mu \geq 0$, giving

$$\begin{aligned} \text{(D)} \quad & \underset{\mu}{\text{maximize}} \quad \mu^T (Ap - b) \\ & \text{subject to} \quad \|A^T \mu\| \leq 1, \\ & \mu \geq 0. \end{aligned} \quad (20)$$

The problem in (20) is therefore the dual problem displayed in the statement.

(b) Let p^* and d^* denote the optimal values of the primal (15) and its dual (20), respectively. The primal objective $\|x - p\|$ is a norm (hence convex) and its constraint $Ax \leq b$ is affine (hence convex). Since the constraint is affine, a Slater point requires no strict inequality and feasibility suffices. Such a point must exist as $\mathcal{P} \neq \emptyset$, so

$$\exists \bar{x} : A\bar{x} \leq b \quad (\text{Slater point}).$$

Moreover, the objective $\|x - p\|$ is bounded below by 0 and the primal is feasible, so $p^* \in \mathbf{R}$. Theorem (Strong Duality) then implies

$$p^* = d^*,$$

and the dual is also solvable since $p^* \in \mathbf{R}$.

Let $\bar{\mu}$ be a dual optimizer. Since $p \notin \mathcal{P}$, the primal value is strictly positive, so

$$\bar{\mu}^T (Ap - b) = d^* = p^* > 0. \quad (21)$$

We can define

$$s := A^T \bar{\mu} \in \mathbf{R}^n, \quad r := b^T \bar{\mu} \in \mathbf{R},$$

and check the three properties of strict separation next.

Consider:

- $s^T p > r$. By the definitions of s and r and the positivity in (21),

$$s^T p - r = (A^T \bar{\mu})^T p - b^T \bar{\mu} = \bar{\mu}^T (Ap - b) > 0.$$

- $s^T x \leq r$. Take any $x \in \mathcal{P}$, so $Ax \leq b$, this is, $b - Ax \geq 0$. With $\bar{\mu} \geq 0$, the entry-wise product gives $\bar{\mu}^T (b - Ax) \geq 0$, so

$$s^T x = (A^T \bar{\mu})^T x = \bar{\mu}^T Ax \leq \bar{\mu}^T b = r.$$

- $s \neq 0$. Suppose, by contradiction, $s = A^T \bar{\mu} = 0$. Take any $\bar{x} \in \mathcal{P}$ (possible as $\mathcal{P} \neq \emptyset$); then $A\bar{x} \leq b$ with $\bar{\mu} \geq 0$ gives $\bar{\mu}^T A\bar{x} \leq \bar{\mu}^T b$, this is, $0 \leq \bar{\mu}^T b$. The dual value at $\bar{\mu}$ would then satisfy

$$d^* = \bar{\mu}^T (Ap - b) = \underbrace{(A^T \bar{\mu})^T p}_{=0} - \bar{\mu}^T b = -\bar{\mu}^T b \leq 0,$$

contradicting (21). Hence $s \neq 0$.

The hyperplane $H_{s,r} = \{x \in \mathbf{R}^n : s^T x = r\}$ therefore separates p strictly from $\mathcal{P}$, in the sense that $s^T p > r$ while $s^T x \leq r$ for every $x \in \mathcal{P}$, as illustrated in Figure 1. 😊

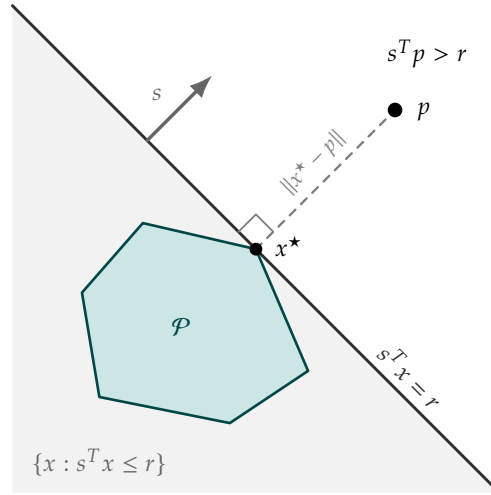


Figure 1: Geometric interpretation.